

Ex. No. 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	
Git Hub Host Link	https://github.com/Karthikeyan-0/flutter.git

Aim

To implement navigation between multiple Flutter screens using navigation and routing techniques.

Description

This experiment demonstrates how to navigate between multiple screens in a Flutter application. Three screens, namely Home, Profile, and Details, are created. `Navigator.push()` and `MaterialPageRoute` are used for screen navigation, while `Navigator.pop()` is used to return to the previous screen. Named routes are also configured using the `routes` property of `MaterialApp`.

Procedure:

1. A new Flutter project named `navigation_app` was created using the Flutter command.
2. Three Dart files named `home.dart`, `profile.dart`, and `details.dart` were created inside the `lib` folder.
3. The Home screen was created with an "Go to Profile" button.
4. `Navigator.push()` with `MaterialPageRoute` was used to navigate from the Home screen to the Profile screen.
5. The Profile screen was created with a "View Details" button.
6. `Navigator.push()` was again used to navigate from the Profile screen to the Details screen.
7. `Navigator.pop()` was implemented in the Details screen to return to the previous screen.
8. Named routes were defined inside `MaterialApp` using the `routes` property for Home, Profile, and Details screens.
9. The application was executed using Flutter and tested in the Chrome browser.
10. The navigation flow was verified as Home → Profile → Details, with back navigation to the previous screen.

Main.dart

```
import 'package:flutter/material.dart';

import 'home.dart';

import 'profile.dart';

import 'details.dart';


void main() {

  runApp(const NavigationApp());

}


class NavigationApp extends StatelessWidget {

  const NavigationApp({super.key});

  @override

  Widget build(BuildContext context) {

    return MaterialApp(

      debugShowCheckedModeBanner: false,

      title: 'Navigation App',

      initialRoute: '/',

      routes: {

        '/': (context) => const HomePage(),

        '/profile': (context) => const ProfilePage(),

        '/details': (context) => const DetailsPage(),
```

```
    },  
  );  
}
```

home.dart

```
import 'package:flutter/material.dart';
```

```
import 'profile.dart';
```

```
class HomePage extends StatelessWidget {
```

```
  const HomePage({super.key});
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return Scaffold(  
      appBar: AppBar(  
        title: const Text('Home'),  
      ),  
      body: Center(  
        child: ElevatedButton(  
          onPressed: () {  
            Navigator.push(  
              context,  
              MaterialPageRoute(  
                builder: (context) => const ProfilePage(),  
              ),  
            );  
          },  
        ),  
      ),  
    );  
  }
```

```
    ),  
    );  
  },  
  child: const Text('Go to Profile'),  
  ),  
),  
);  
}  
}
```

profile.dart

```
import 'package:flutter/material.dart';  
  
import 'details.dart';  
  
class ProfilePage extends StatelessWidget {  
  const ProfilePage({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: AppBar(  
        title: const Text('Profile'),  
      ),  
      body: Center(  
        child: ElevatedButton(  

```

```
onPressed: () {  
  
  Navigator.push(  
  
    context,  
  
    MaterialPageRoute(  
  
      builder: (context) => const DetailsPage(),  
  
    ),  
  
  );  
  
  },  
  
  child: const Text('View Details'),  
  
  ),  
  
  ),  
  
);  
  
}  
  
}
```

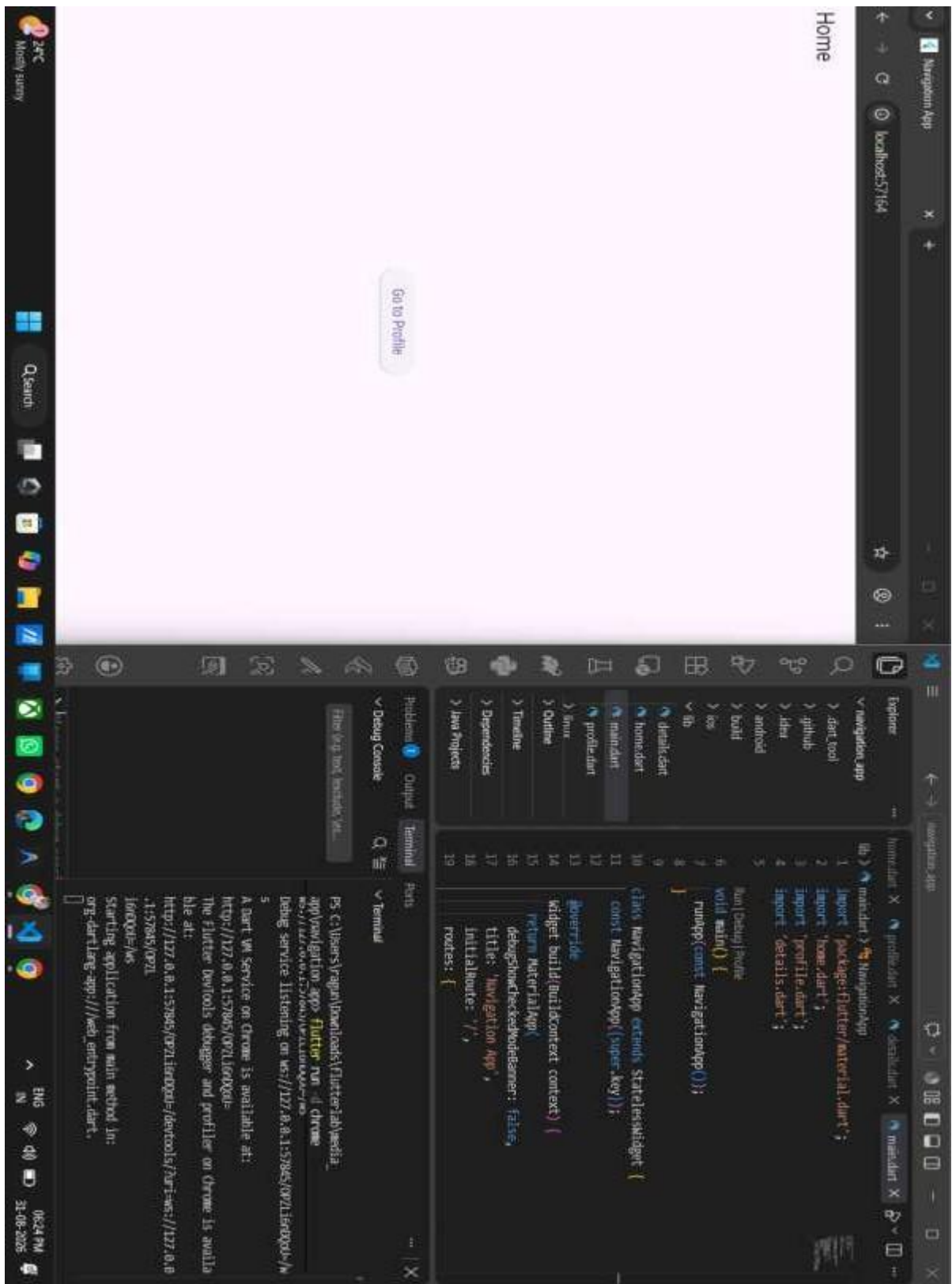
details.dart

```
import 'package:flutter/material.dart';  
  
class DetailsPage extends StatelessWidget {  
  
  const DetailsPage({super.key});  
  
  @override  
  
  Widget build(BuildContext context) {  
  
    return Scaffold(  

```

```
appBar: AppBar(  
  title: const Text('Details'),  
),  
body: Center(  
  child: ElevatedButton(  
    onPressed: () {  
      Navigator.pop(context);  
    },  
    child: const Text('Back'),  
  ),  
),  
);  
}  
}
```

Output



Result

Thus, the Flutter application was successfully developed to implement navigation and routing between multiple screens. The application successfully navigated from Home → Profile → Details and allowed the user to return to the previous screen using back navigation.